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Informing net-zero policies in the European Union with evidence-based approaches

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Leveraging experimental methods, such as randomized controlled trials, can inform energy and climate policy delivery in the European Union and maximize its effectiveness.

The European Union (EU) has set decarbonization goals such as boosting renewable energy, improving energy efficiency and investing in clean transport through the European Green Deal and the 'Fit for 55' legislation[1]. Reaching these goals will require the design of robust policies and the monitoring of their implementation. However, there has so far largely been a lack of rigorous experiments with a credible counterfactual for such policies, which are necessary to identify the causal impact of a given policy. A credible counterfactual refers to a plausible alternative scenario describing what would have happened in the absence of a specific intervention, event or treatment. It serves as a realistic benchmark for comparison, enabling evaluators or researchers to determine the causal effect of an intervention or decision. Without a credible counterfactual, policymakers cannot be sure that policies achieve their intended effects on specific outcomes, such as carbon emission reductions. Indeed, some evidence indicates that these outcomes are often not achieved: among 1,500 climate policies, only 63 could be credibly linked to sizeable emissions reductions[2].

Evidence-based approaches could help. Experimental methods have become a useful tool to rigorously evaluate the causal effect of policies. As described in the US Department of Energy's evaluation plans[1], experimental and quasi-experimental methods have a critical role in identifying the causal effects of programmes and improving decision-making processes. Applying an experimental approach to clean technology policies could yield returns on investment ranging from 21% to 37%[3].

These approaches are used in policy-making elsewhere, such as in the USA with the adoption of the Foundations for Evidence-Based Policymaking Act (2018). However, EU regulation has yet to fully embrace evidence-based policy approaches[4]. In this Comment, we discuss the added value of incorporating experimental methods[5] such as randomized controlled trials (RCTs) into upcoming EU legislation on clean technology and energy policies (Fig. 1).

Experimental methods for policy-making

Economic field experiments (that is, RCTs) and quasi-experimental methods (such as instrumental variable design, regression discontinuity design or difference-in-differences design) can be used in policy design to understand the effect of the policy. They use counterfactuals to observe the difference between what would happen with and without a given policy. Field experiments (RCTs) are the more robust of these methods, as they do not rely on as many assumptions as quasi-experimental methods[5]. They are therefore able to disentangle the effect of a policy from potentially confounding factors that might also influence the outcome and provide proof of concept (that is, proof that a policy has the desired effect). Proof of concept provides more certainty around policy outcomes, especially when their potential usefulness is contested. For example, rigorous evidence on industrial policies has helped to understand the benefits of their intended results[6].

Of experimental methods, RCTs — experiments in which participants are randomly assigned to either a treatment or a control group to measure the causal impact of an intervention — deliver the highest degree of confidence in the empirical results if the sample is balanced[7]. Furthermore, the researcher has full control over the research design, allowing them to answer exactly the question they desire.

To provide confidence to decision makers, policy-making needs to be done strategically and allow for experimentation. In particular, RCTs require a high number of units of observation to allow for randomization[1]. Therefore, we recommend that RCTs take an iterative approach, starting with a small pilot experiment, the results of which inform larger, more ambitious trials and fine-tuning of policy modalities. Smaller experiments save time and money, minimize disruptions and can be conducted faster, and create buy-in for the novel approach, visibly demonstrate the benefits to the relevant stakeholders in public institutions and minimize potential harm if things go not as planned.

Importantly for energy and climate policy, experiments can not only help to determine what works, thus leading to more efficient use of resources, but can also inform policymakers on the underlying mechanisms as to why policies operate in the way that they do[8]. Uncovering mechanisms in economic research using RCTs is important because it helps researchers to understand why and how an intervention works (or not), allowing for improved theory building, more targeted policies, generalizability of results and informed scaling decisions — all highly relevant for energy and climate policies as they will likely increase in relevance and funding in the near future. For instance, an experimental approach to financial incentives could provide evidence that sheds light on existing unknowns in climate policies[2], such as whether these types of incentive are able to increase adoption of renewable energy, especially compared with other policy approaches (such as information provision)[9].

RCTs and European legislation

There is no comprehensive EU approach for rigorous evidence-based decision-making for informing net-zero policies. In the absence of rigorous evidence at the European level, policymakers cannot credibly assess the effectiveness of their programmes and policies. As an RCT of the US Weatherization Assistance Program demonstrated, even the

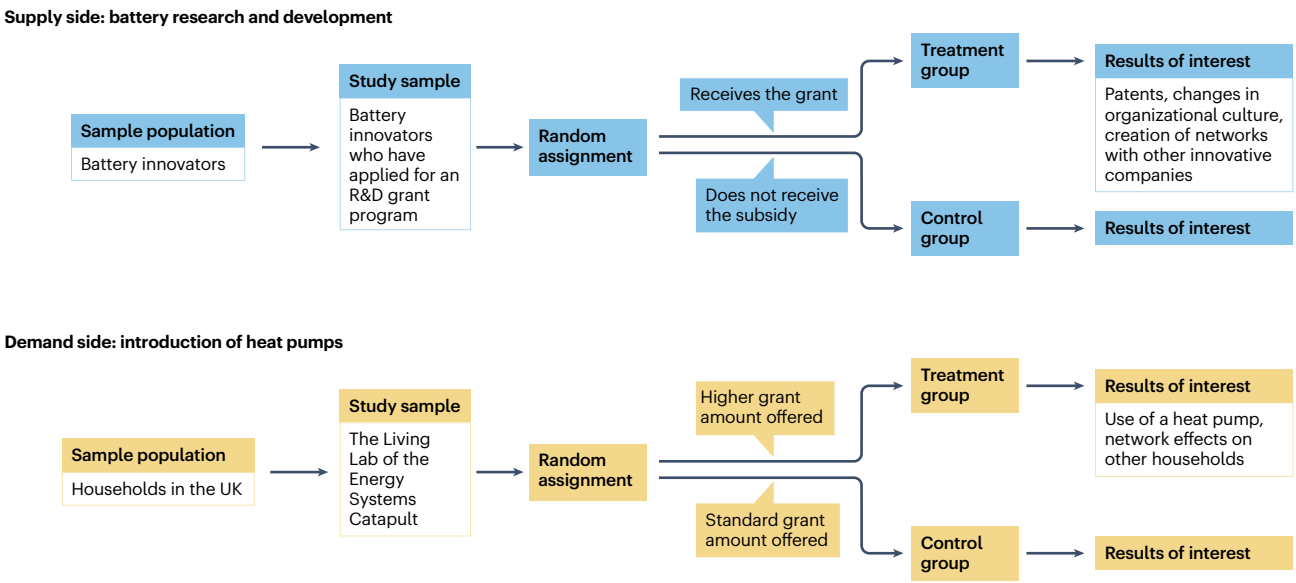


Fig. 1 | Randomized controlled trials in clean technology policies. Randomized controlled trials can be used to understand effectiveness and impacts of supply-side and demand-side policies. R&D, research and development.

most well-intentioned programmes, at a large scale, can fail to deliver[9]. Untested policies are a gamble at a time when climate change is affecting many parts of the world. Therefore, policymakers need a reliable roadmap for action.

EU policymakers should integrate experimentation into policies and programmes under the European Green Deal and the Fit for 55 agenda[10] and establish an institutionalized framework for continuous experimentation and use of evidence. The first step could be setting a learning agenda to identify pressing knowledge gaps for which more evidence needs to be generated. For example, for most emission-reduction policies, no causal evidence exists that directly links the policy to emission reductions[11]. Even though not all policies can be rigorously evaluated (for example, because large-scale infrastructure policies do not allow for randomization), many opportunities exist, for example in the area of energy efficiency, technology adoption and use[9,11], induced learning curves and behavioural and technological spillovers.

For RCTs to be successfully implemented in EU policy-making, key stakeholders (such as policymakers, research institutions and industry leaders) need to embrace a novel approach of co-creating evidence and a 'learning culture' that allows for a policy to be found ineffective and hence discontinued. Such approaches are already being implemented in specific best practice examples, such as the Energy and Environment Lab at the University of Chicago or the Centre for Net Zero in the UK.

Leveraging the EU's networks would allow for comprehensive and impactful experimentation[12] across member states, accelerating the adoption of successful green policies. A centralized coordination body could facilitate the integration of RCTs into policy design, ensuring that experiments are planned from the outset. Additionally, building permanent experimentation capacities within EU and national institutions could support the systematic use of RCTs.

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Competing interests

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